

White Paper 5 — Binding-pocket detection & structural druggability

Pipeline step: Binding-pocket detection & structural druggability **Inputs:** lead structures (apo), reference ligand positions
Outputs: `pocket_report.csv`, `pocket_druggability_summary.csv`, `fig_pockets_druggability.png`

1. Purpose

Knowledge-base tractability (White Paper 2) tells us whether a target has *attracted* chemistry; structural druggability tells us whether its pocket is *physically capable* of binding a drug-like molecule. This step detects pockets on each lead structure and scores their druggability directly from geometry and chemistry — the structural counterpart to the annotation-based assessment.

2. The method: fpocket

Pocket detection used **fpocket**, an open-source geometry-based cavity detector ^[1]. fpocket places "alpha spheres" that fit between protein atoms; clusters of appropriately-sized spheres define candidate pockets. Critically, fpocket reports a **druggability score** (0–1) trained to discriminate pockets that bind drug-like ligands from those that do not, using descriptors such as pocket volume, hydrophobicity, and enclosure. This score is the structural analog of the Open Targets tractability bucket.

3. Preparation and pocket matching

Each structure was stripped of its ligand and waters to an **apo** form (STAT3 4,172, RORC 2,179, JAK3 2,333 protein atoms) before pocket detection, so that fpocket found the pocket *de novo* rather than around the ligand. To identify which detected pocket corresponds to the functional site, we matched pockets to the known crystallographic ligand position by **ligand-atom overlap** (protein pocket atoms within 4.5 Å of the reference ligand) — more robust than centroid distance when a protein has several nearby cavities.

4. Results and interpretation

Lead	Matched pocket	fpocket druggability	Volume (Å ³)	Interpretation
RORC / RORyt	LBD orthosteric	0.936	1,124	Textbook druggable pocket
JAK3	ATP site	0.016	1,860	Chemically validated despite low geometric score
STAT3	SH2 / core	0.001	199	Genuinely shallow — a hard TF target

Three distinct situations emerge, and reading them correctly is the point of the step:

- **RORyt** scores 0.94 — a deep, enclosed, hydrophobic ligand-binding domain pocket. Structural and chemical druggability agree, making it the natural primary docking target.
- **JAK3** scores low *in this crystal form* even though its ATP site is one of the most drugged pockets in medicine (6,069 potent inhibitors). fpocket's geometric score is sensitive to the particular conformation and the bound-ligand-induced cleft; the discrepancy is a caution against reading fpocket in isolation, not evidence of undruggability.
- **STAT3** scores ~0.001 — the SH2/core surface is genuinely shallow and solvent-exposed. This is the structural reason the field has moved to degrader and bivalent chemistry for STAT3, consistent with the degrader-class ligand in the chosen structure. The low score is a true finding, not an artifact.

The privileged-pocket logic underlying this interpretation is the druggable-genome framework ^[2].

5. Docking-box definition

For each lead the docking search center was set from the **experimental ligand position** rather than the fpocket centroid, because the crystallographic ligand is the most reliable marker of the functional site. These centers feed the docking step (White Paper 7).

6. Deliverables

- `pocket_report.csv` — every detected pocket with druggability, volume, and hydrophobicity.
- `pocket_druggability_summary.csv` — the matched functional pocket per lead.

- `fig_pockets_druggability.png` — 3D pocket views plus a druggability-comparison panel.

7. Caveats

- fpocket scores are conformation-dependent; a single crystal form can under- or over-score a pocket (JAK3 is the clear example here).
- Structural druggability and chemical tractability are complementary — neither alone is decisive.

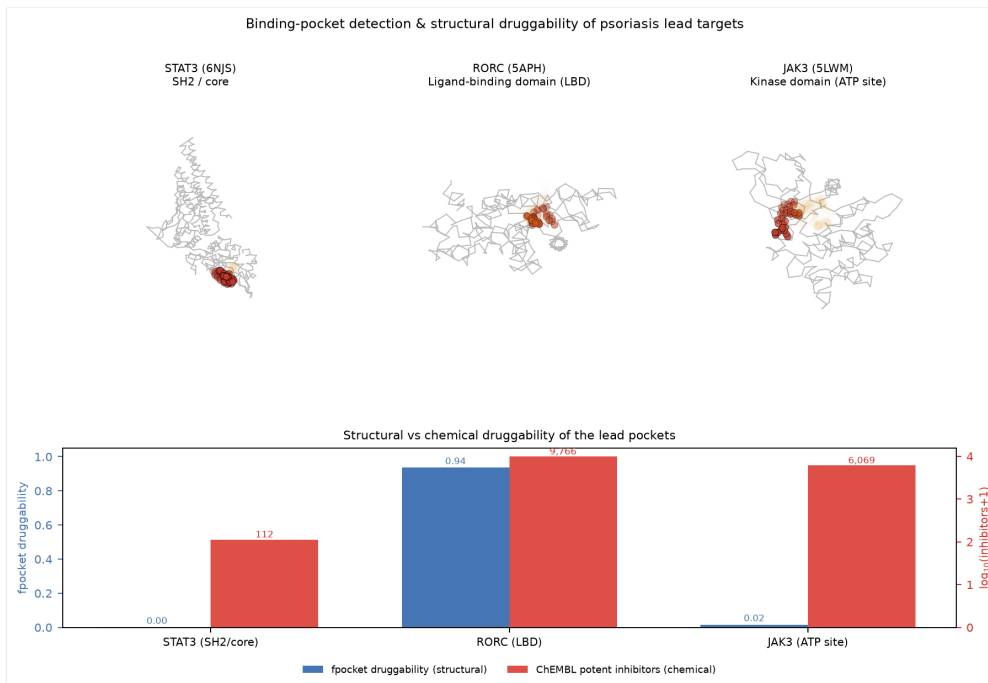


Figure: key output of this pipeline step.

References

1. Le Guilloux et al. Fpocket: An open source platform for ligand pocket detection. *BMC Bioinformatics*. 2009. doi:[10.1186/1471-2105-10-168](https://doi.org/10.1186/1471-2105-10-168)
2. Hopkins et al. The druggable genome. *Nature Reviews Drug Discovery*. 2002. doi:[10.1038/nrd892](https://doi.org/10.1038/nrd892)